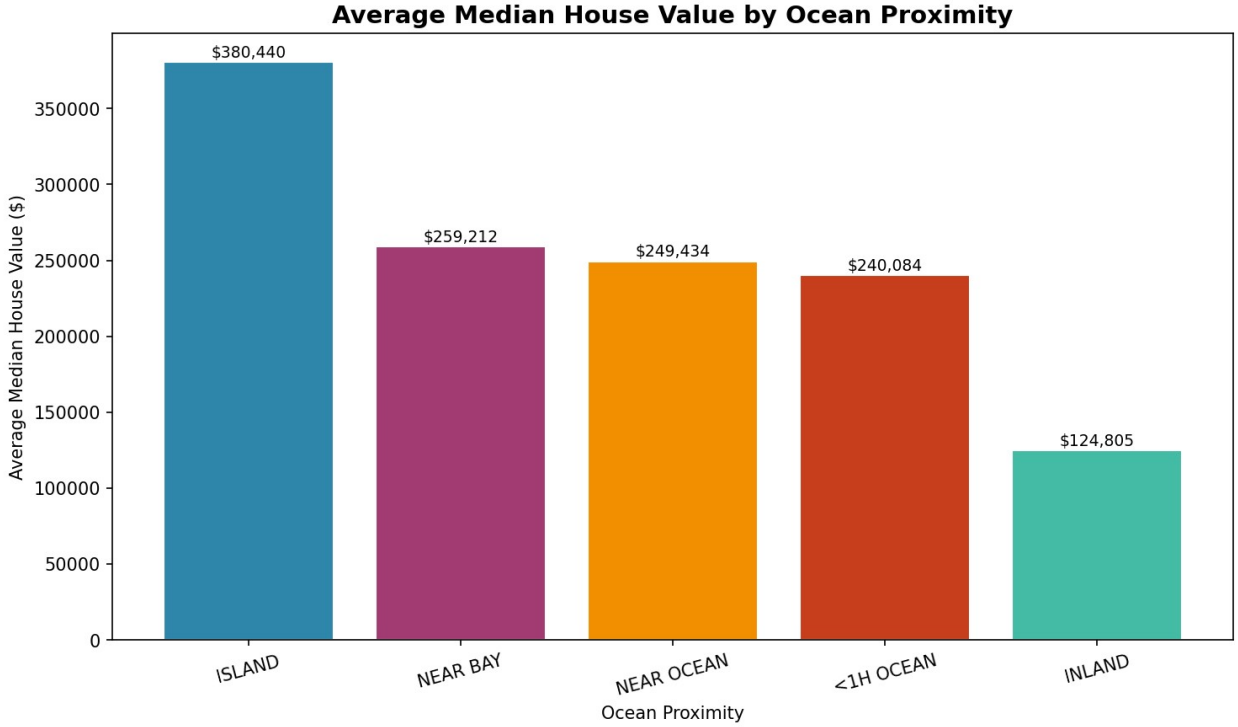
L1F21BSCS0097

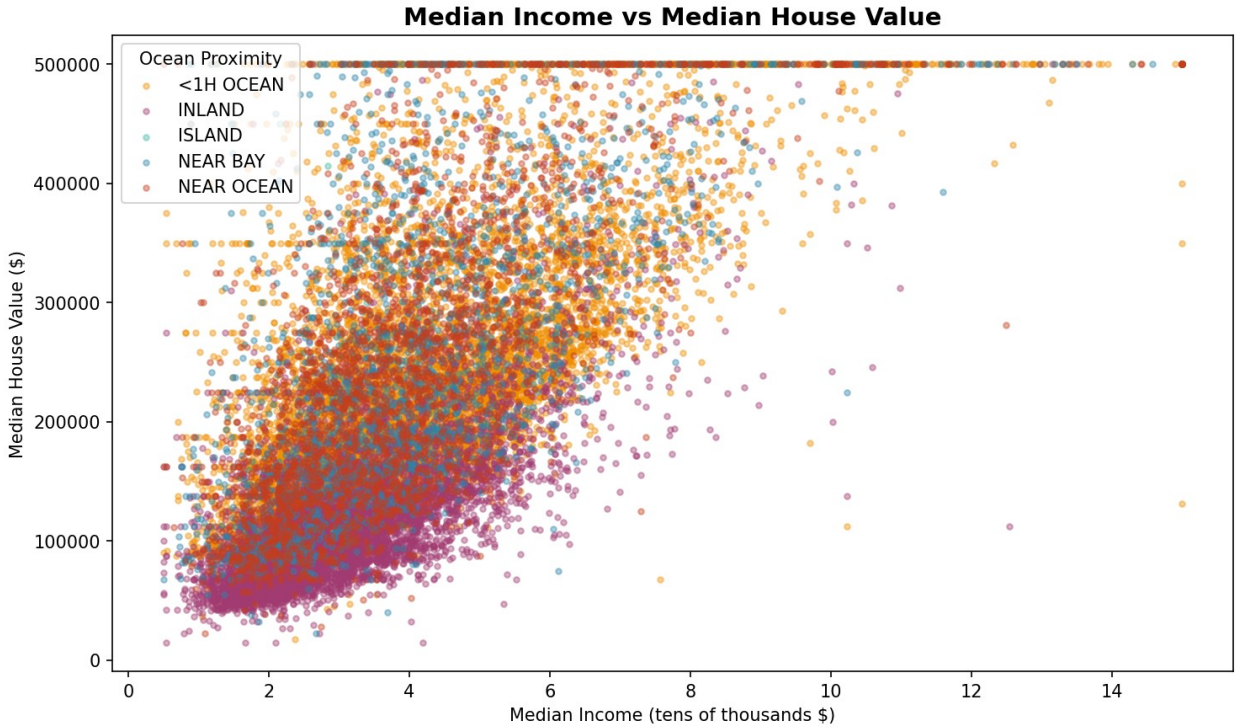
Ahmad Sachyar

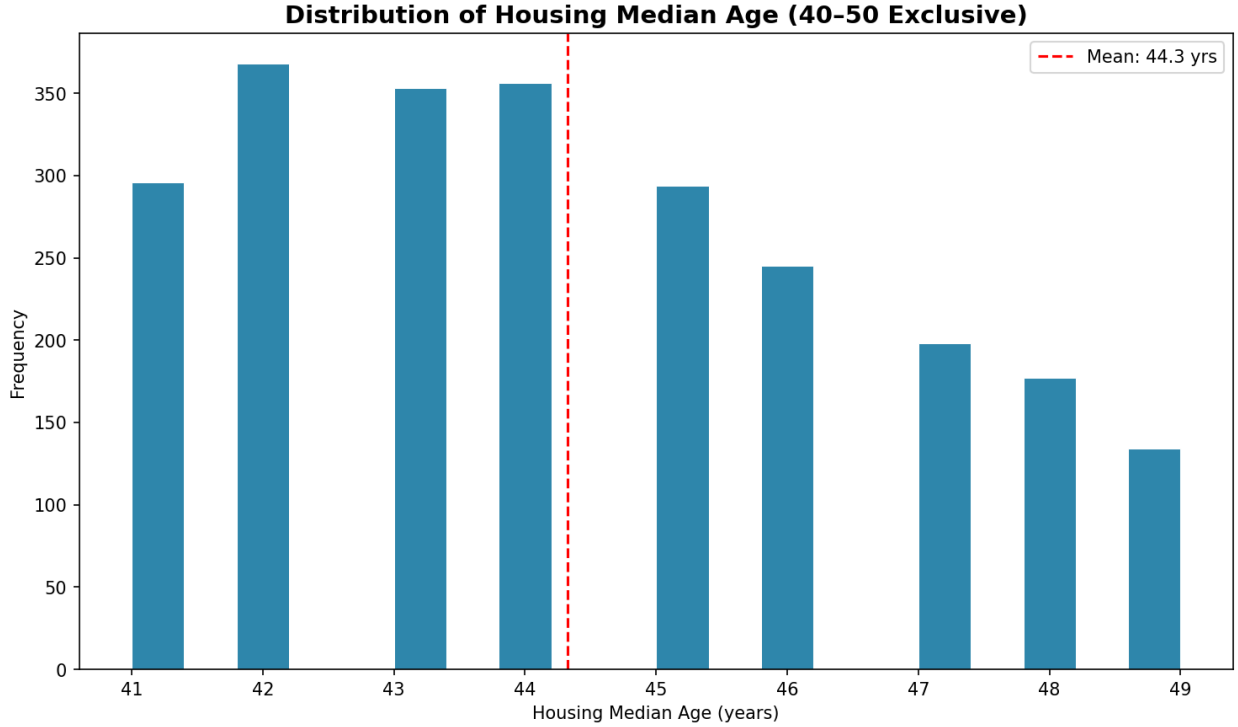
Assignment 1

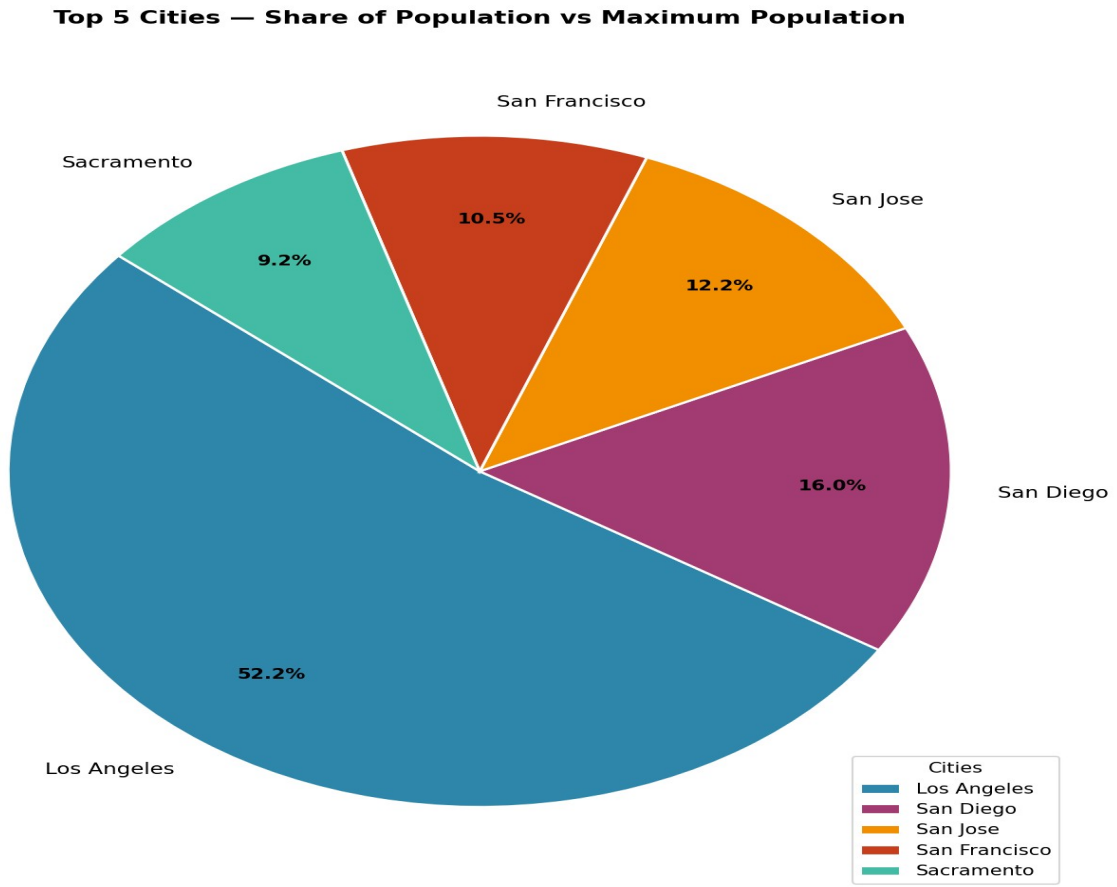
Submitted to:

Sir Ali Raza


Average Median House Value by Ocean Proximity
$380,440
$259,212
$249,434
$240,084
$124,805
Average Median House Value ($)
0
50000
100000
150000
200000
250000
300000
350000
ISLAND
NEAR BAY
NEAR OCEAN
<1H OCEAN
INLAND
Ocean Proximity


Median Income vs Median House Value
Ocean Proximity
<1H OCEAN
INLAND
ISLAND
NEAR BAY
NEAR OCEAN
Median House Value ($)
0
100000
200000
300000
400000
500000
0
2
4
6
8
10
12
14
Median Income (tens of thousands $)


Distribution of Housing Median Age (40–50 Exclusive)
Mean: 44.3 yrs
Frequency
0
50
100
150
200
250
300
350
41
42
43
44
45
46
47
48
49
Housing Median Age (years)


Top 5 Cities — Share of Population vs Maximum Population
San Francisco
Sacramento
San Jose
10.5%
9.2%
12.2%
16.0%
San Diego
52.2%
Los Angeles
Cities
Los Angeles
San Diego
San Jose
San Francisco
Sacramento

```python
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
print("L1F21BSCS0097 Ahmad Sachyar")
zipcodevalues = ["94705.0", "94611.0", "94618.0", "95935.0", "95972.0",
"95962.0"]
housing_data      = pd.read_csv("housing.csv")
housing_addresses = pd.read_csv("housing_addresses.csv")
print("Task 1")
print("housing_data")
print(housing_data.head(10).to_string(index=True))
print()
print("housing_addresses")
print(housing_addresses.head(10).to_string(index=True))
print()
print("L1F21BSCS0097 Ahmad Sachyar")
print("Task 2")
print("housing_addresses")
print()
housing_addresses.rename(columns={"postal_code": "zip_code"}, inplace=True)
print(housing_addresses.head(1).to_string(index=True))
housing_addresses.rename(columns={"zip_code" : "postal_code"}, inplace=True)
print()
print("L1F21BSCS0097 Ahmad Sachyar")
print("Task 3")
print("housing_data")
print()
housing_data["zip_code"] = None
print(housing_data.head(1).to_string(index=True))
print()
print("L1F21BSCS0097 Ahmad Sachyar")
print("Task 4")
print("housing_data")
print()
housing_data["zip_code"] = np.random.choice(zipcodevalues,
size=len(housing_data))
print(housing_data.head(5).to_string(index=True))
print("Task 5")
print("L1F21BSCS0097 Ahmad Sachyar")
print("Data Frame L1F21BSCS0097")
print()
housing_addresses[['lat_key', 'lon_key']] = (
    housing_addresses['latitude_longitude']
    .str.split(',', expand=True)
    .astype(float)
)
housing_data['lat_key']      = housing_data['latitude']
housing_data['lon_key']      = housing_data['longitude']
housing_addresses['lat_key'] = housing_addresses['lat_key']
housing_addresses['lon_key'] = housing_addresses['lon_key']
L1F21BSCS0097 = pd.merge(
    housing_data,
    housing_addresses,
    on=['lat_key', 'lon_key'],
    how='left'
```

```
)
L1F21BSCS0097.drop(columns=['lat_key', 'lon_key'], inplace=True)
print(L1F21BSCS0097.head(10).to_string(index=True))
print()
print("Task 5")
print()
print("Bar Chart")
avg_value = L1F21BSCS0097.groupby('ocean_proximity')
['median_house_value'].mean().sort_values(ascending=False)
plt.figure(figsize=(10, 6))
bars = plt.bar(avg_value.index, avg_value.values,
color=['#2E86AB','#A23B72','#F18F01','#C73E1D','#44BBA4'], edgecolor='white')
plt.title('Average Median House Value by Ocean Proximity', fontsize=14,
fontweight='bold')
plt.xlabel('Ocean Proximity')
plt.ylabel('Average Median House Value ($)')
plt.xticks(rotation=15)
for bar, val in zip(bars, avg_value.values):
    plt.text(bar.get_x() + bar.get_width()/2, bar.get_height() + 1000,
             f'${val:,.0f}', ha='center', va='bottom', fontsize=9)
plt.tight_layout()
plt.savefig('barchart.png', dpi=150, bbox_inches='tight')
plt.show()
print()
print("Scatter Plot")
colors_map = {
    'NEAR BAY':   '#2E86AB',
    'INLAND':     '#A23B72',
    '<1H OCEAN':  '#F18F01',
    'NEAR OCEAN': '#C73E1D',
    'ISLAND':     '#44BBA4'
}
plt.figure(figsize=(10, 6))
for proximity, group in L1F21BSCS0097.groupby('ocean_proximity'):
    plt.scatter(group['median_income'],
                group['median_house_value'],
                label=proximity,
                color=colors_map.get(proximity, 'gray'),
                alpha=0.4,
                s=10)

plt.title('Median Income vs Median House Value', fontsize=14, fontweight='bold')
plt.xlabel('Median Income (tens of thousands $)')
plt.ylabel('Median House Value ($)')
plt.legend(title='Ocean Proximity', loc='upper left')
plt.tight_layout()
plt.savefig('scatterplot.png', dpi=150, bbox_inches='tight')
plt.show()
print()
print("Histogram")
filtered_age = L1F21BSCS0097[
    (L1F21BSCS0097['housing_median_age'] > 40) &
    (L1F21BSCS0097['housing_median_age'] < 50)
]['housing_median_age']
```

```python
plt.figure(figsize=(10, 6))
plt.hist(filtered_age, bins=20, color='#2E86AB', edgecolor='white',
linewidth=0.5)
plt.title('Distribution of Housing Median Age (40–50 Exclusive)', fontsize=14,
fontweight='bold')
plt.xlabel('Housing Median Age (years)')
plt.ylabel('Frequency')
plt.axvline(filtered_age.mean(), color='red', linestyle='--', linewidth=1.5,
            label=f'Mean: {filtered_age.mean():.1f} yrs')
plt.legend()
plt.tight_layout()
plt.savefig('histogram.png', dpi=150, bbox_inches='tight')
plt.show()
print()
print("Pie Chart")
city_population = L1F21BSCS0097.groupby('locality-political')['population'].sum()
top5_cities     = city_population.sort_values(ascending=False).head(5)
max_population  = city_population.max()

top5_share = (top5_cities / max_population) * 100

plt.figure(figsize=(9, 9))
wedges, texts, autotexts = plt.pie(
    top5_share.values,
    labels=top5_share.index,
    autopct='%1.1f%%',
    colors=['#2E86AB','#A23B72','#F18F01','#C73E1D','#44BBA4'],
    startangle=140,
    pctdistance=0.75,
    wedgeprops=dict(edgecolor='white', linewidth=1.5)
)
for text in texts:
    text.set_fontsize(11)
for autotext in autotexts:
    autotext.set_fontsize(10)
    autotext.set_fontweight('bold')

plt.title('Top 5 Cities — Share of Population vs Maximum Population',
          fontsize=13, fontweight='bold', pad=20)
plt.legend(wedges, top5_share.index, title='Cities', loc='lower right',
fontsize=10)
plt.tight_layout()
plt.savefig('piechart.png', dpi=150, bbox_inches='tight')
plt.show()
print("L1F21BSCS0097 Ahmad Sachyar")
```

```
Running....

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Task 1
housing_data
   longitude  latitude  housing_median_age  total_rooms  total_bedrooms  population  households  median_income  median_house_value ocean_proximity
0    -122.23     37.88                41.0        880.0           129.0       322.0       126.0         8.3252            452600.0        NEAR BAY
1    -122.22     37.86                21.0       7099.0          1106.0      2401.0      1138.0         8.3014            358500.0        NEAR BAY
2    -122.24     37.85                52.0       1467.0           190.0       496.0       177.0         7.2574            352100.0        NEAR BAY
3    -122.25     37.85                52.0       1274.0           235.0       558.0       219.0         5.6431            341300.0        NEAR BAY
4    -122.25     37.85                52.0       1627.0           280.0       565.0       259.0         3.8462            342200.0        NEAR BAY
5    -122.25     37.85                52.0        919.0           213.0       413.0       193.0         4.0368            269700.0        NEAR BAY
6    -122.25     37.84                52.0       2535.0           489.0      1094.0       514.0         3.6591            299200.0        NEAR BAY
7    -122.25     37.84                52.0       3104.0           687.0      1157.0       647.0         3.1200            241400.0        NEAR BAY
8    -122.26     37.84                42.0       2555.0           665.0      1206.0       595.0         2.0804            226700.0        NEAR BAY
9    -122.25     37.84                52.0       3549.0           707.0      1551.0       714.0         3.6912            261100.0        NEAR BAY

housing_addresses
  street_number                  route locality-political  postal_code latitude_longitude
0          3130  Grizzly Peak Boulevard           Berkeley      94705.0      37.88,-122.23
1          2005             Tunnel Road            Oakland      94611.0      37.86,-122.22
2          6886             Chabot Road            Oakland      94618.0      37.85,-122.24
3          6365           Florio Street            Oakland      94618.0      37.85,-122.25
4          5407           Bryant Avenue            Oakland      94618.0      37.84,-122.25
5          5321        Claremont Avenue            Oakland      94618.0      37.84,-122.26
6          6398        Telegraph Avenue            Oakland      94609.0      37.85,-122.26
7          1889           Harmon Street           Berkeley      94703.0      37.85,-122.27
8           812             55th Street            Oakland      94608.0      37.84,-122.27
9          1408             66th Street           Berkeley      94702.0      37.85,-122.28
```

```
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Task 2
housing_addresses

  street_number                  route locality-political  zip_code latitude_longitude
0          3130  Grizzly Peak Boulevard           Berkeley   94705.0      37.88,-122.23

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Task 3
housing_data

   longitude  latitude  housing_median_age  total_rooms  total_bedrooms  population  households  median_income  median_house_value ocean_proximity zip_code
0    -122.23     37.88                41.0        880.0           129.0       322.0       126.0         8.3252            452600.0        NEAR BAY     None
```

```
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Task 4
housing_data

   longitude  latitude  housing_median_age  total_rooms  total_bedrooms  population  households  median_income  median_house_value ocean_proximity zip_code
0    -122.23     37.88                41.0        880.0           129.0       322.0       126.0         8.3252            452600.0        NEAR BAY  95962.0
1    -122.22     37.86                21.0       7099.0          1106.0      2401.0      1138.0         8.3014            358500.0        NEAR BAY  95935.0
2    -122.24     37.85                52.0       1467.0           190.0       496.0       177.0         7.2574            352100.0        NEAR BAY  95972.0
3    -122.25     37.85                52.0       1274.0           235.0       558.0       219.0         5.6431            341300.0        NEAR BAY  95972.0
4    -122.25     37.85                52.0       1627.0           280.0       565.0       259.0         3.8462            342200.0        NEAR BAY  94618.0
Task 5
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Data Frame L1F21BSCS0097

   longitude  latitude  housing_median_age  total_rooms  total_bedrooms  population  households  median_income  median_house_value ocean_proximity zip_code street_number                  route locality-political  postal_code latitude_longitude
0    -122.23     37.88                41.0        880.0           129.0       322.0       126.0         8.3252            452600.0        NEAR BAY  95962.0         3130  Grizzly Peak Boulevard           Berkeley      94705.0      37.88,-122.23
1    -122.22     37.86                21.0       7099.0          1106.0      2401.0      1138.0         8.3014            358500.0        NEAR BAY  95935.0         2005             Tunnel Road            Oakland      94611.0      37.86,-122.22
2    -122.24     37.85                52.0       1467.0           190.0       496.0       177.0         7.2574            352100.0        NEAR BAY  95972.0         6886             Chabot Road            Oakland      94618.0      37.85,-122.24
3    -122.25     37.85                52.0       1274.0           235.0       558.0       219.0         5.6431            341300.0        NEAR BAY  95972.0         6365           Florio Street            Oakland      94618.0      37.85,-122.25
4    -122.25     37.85                52.0       1627.0           280.0       565.0       259.0         3.8462            342200.0        NEAR BAY  94618.0         6365           Florio Street            Oakland      94618.0      37.85,-122.25
5    -122.25     37.85                52.0        919.0           213.0       413.0       193.0         4.0368            269700.0        NEAR BAY  95962.0         6365           Florio Street            Oakland      94618.0      37.85,-122.25
6    -122.25     37.84                52.0       2535.0           489.0      1094.0       514.0         3.6591            299200.0        NEAR BAY  94611.0         5407           Bryant Avenue            Oakland      94618.0      37.84,-122.25
7    -122.25     37.84                52.0       3104.0           687.0      1157.0       647.0         3.1200            241400.0        NEAR BAY  94618.0         5407           Bryant Avenue            Oakland      94618.0      37.84,-122.25
8    -122.26     37.84                42.0       2555.0           665.0      1206.0       595.0         2.0804            226700.0        NEAR BAY  94705.0         5321        Claremont Avenue            Oakland      94618.0      37.84,-122.26
9    -122.25     37.84                52.0       3549.0           707.0      1551.0       714.0         3.6912            261100.0        NEAR BAY  94705.0         5407           Bryant Avenue            Oakland      94618.0      37.84,-122.25

Task 6

Bar Chart

Scatter Plot

Histogram

Pie Chart
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